2024秋线性代数期中(回忆版)

1. Multiple Choice. Only one choice is correct.

(1) Suppose A is an $m\times n$ real matrix with $m<n$, and the equation $Ax=b$ has a solution for any m-dimensional real column vector b. Which of the following assertions must be true?

(A) $Ax=b$ has a unique solution for every $b\in R^m$.

(B) $A^Tx=d$ has a solution for any $d\in R^n$.

(C) $N(A)=\{0\}$.

(D) A has a right inverse.

(2) Suppose we have matrices A,B satisfying $EA=B$ for some invertible matrix E. Which of the following assertions must be true?

(A) $N(A)=N(B)$ (B) $C(A)=C(B)$

(C) $N(A^T)=N(B^T)$ (D) $A^TA=B^TB$

(3) For any $m\times n$ matrix A with reduced row echelon from U. Which of the following assertions must be true?

(A) $C(A)=C(A^TA)$ (B) $\text{rank}(A)=\text{rank}(A^TA)$

(C) $C(A)=C(U)$ (D) If $C(A)=R^m$, then A^TA is invertible.

(4) Let A,B,C be $n\times n$ matrices such that $ABC=I$, where I is the identity matrix of order n, then

(A) $ACB=I$ (B) $CBA=I$ (C) $BCA=I$ (D) $BAC=I$

(5) Let A be an $n\times n$ matrix $(n>1)$ such that $A=A^2$, and I be the $n\times n$ identity matrix, then

(A) $\text{rank}(A)+\text{rank}(A-I)>n$ (B) $\text{rank}(A)+\text{rank}(A-I)<n$

(C) $\text{rank}(A)+\text{rank}(A-I)=n$ (D) $\text{rank}(A)+\text{rank}(A-I)=n-1$.

2. Fill in the blanks.

(1) Let $A=\begin{bmatrix}1&1&1\\0&2&1\\0&0&3\end{bmatrix}$, then $A^{-1}=$_____.

(2) Let $A=\begin{bmatrix}1&2&3\\2&-1&1\\3&1&3\end{bmatrix}$. The LDU factorization of A has $L=$_____.

(3) The matrix which projects every vector b in R^3 onto the line in the direction of $\begin{bmatrix}2\\3\\1\end{bmatrix}$ through the origin is _____.

(4) Let A be a 2024×2025 real matrix with $\dim N(A)=11$, then $\dim N(A^T)=$_____.

3. Let $A=\begin{bmatrix}0&1&0\\1&0&-1\\0&1&0\end{bmatrix}$

Consider $X-XA^2-AX+AXA^2=I$,

where I is the 3×3 identity matrix, and X is a 3×3 matrix.

(a) Compute $I-A$ and $I-A^2$.

(b) Find all possible X.

4. Let $A=\begin{bmatrix}1&0&2&0\\1&1&3&0\\1&0&2&1\\3&2&8&0\end{bmatrix}$, $b=\begin{bmatrix}3\\3\\3\\9\end{bmatrix}$

(a) Find the reduced row echelon ~~from~~ form of A.

(b) Find a basis for the row space $C(A^T)$, the column space $C(A)$, and the left nullspace $N(A^T)$.

(c) Find the complete solution to $Ax=b$. In other words, find all the solutions to $Ax=b$.

5. Consider the following subspace of R^3:

$$V=\left\{\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}\in R^3 \mid x_1+x_2+x_3=0\right\}$$

(a) show that: $V_1=\begin{bmatrix}-1\\1\\0\end{bmatrix}$, $V_2=\begin{bmatrix}-1\\0\\1\end{bmatrix}$ is a basis of V.

(b) Let T be the linear transformation from V to R^3 defined as follows:

$$T\left(\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}\right)=\begin{bmatrix}2x_1-x_2\\x_2+x_3\\x_1\end{bmatrix}$$

Find the matrix reprosentation of T with respect to the ordered basis v_1, v_2 of V and the ordered basis

$$e_1=\begin{bmatrix}1\\0\\0\end{bmatrix}, e_2=\begin{bmatrix}0\\1\\0\end{bmatrix}, e_3=\begin{bmatrix}0\\0\\1\end{bmatrix} \text{ of } R^3.$$

(c) Can we find a voctor $v\in V$ such that $T(v)=\begin{bmatrix}3\\2\\1\end{bmatrix}$? If so, find one such v. Otherwise give an explanation.

6. Let $A=\begin{bmatrix}1&1&1\\0&0&1\\0&0&-1\end{bmatrix}$ and $U=\{B\in M_{3\times3}(R) \mid AB-BA\}$,

where $M_{3\times3}(R)$ denotes the vector space of all 3×3 real matrices with the ordinary matrix addition and scalar multiplication.

(a) Show that U is a subspace of $M_{3\times3}(R)$.

(b) Find a basis of U.

(c) Find the dimension of U.

7. Prove the following two independent statements.

(a) Let A be an $m\times n$ real matrix. Suppose $v_1, v_2, \cdots, v_n$ is a basis of $C(A^T)$ and $w_1, w_2, \cdots, w_t$ is a basis of $N(A)$. Show that : $s+t=n$ and

$$v_1, v_2, \cdots, v_n, w_1, w_2, \cdots, w_t.$$

is a basis of R^n.

(b) Let A be an $m\times n$ real matrix with rank m. Show that AA^T is invertible.